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Course ID: STA 201

Course title: Elements of Statistics and Probability

Section: 06

Answer to the Question No-1

(a)

My student id : 24101226 , $X = 2$, $Y = 6$

Marks	Frequency	
	Male	Female
0-10	8	5
10-20	17	22
20-30	33	26
30-40	22	33
40-50	10	15
Total	90	101

(b) For male,

Class Interval
(c)

Class Interval (c)	Mid-point (x_i)	Frequency (f_i)	Cumulative Frequency (F)	$f_i x_i$
0-10	5	8	8	40
10-20	15	17	25	255
20-30	25	33	58	825
30-40	35	22	80	770
40-50	45	10	90	450
Total		90		2340

$$\text{Arithmetic Mean, } \bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{2340}{90} = 26$$

$$\text{Median, } M_e = L_o + \frac{\frac{n}{2} - F_{-1}}{f_m} \times C$$

$$= 20 + \frac{45 - 25}{42} \times 10$$

$$\approx 24.762$$

Median class = $\frac{n}{2}$ ^{45th} observation
= "20-30" class

$$L_o = 20, n = 90, F_{-1} = 25, f_m = 42,$$

$$C = 10$$

$$\text{Mode, } M_o = L_o + \frac{(f_o - f_{-1})}{(f_o - f_{-1}) + (f_o - f_1)} \times C$$

$$= 20 + \frac{(33 - 17)}{(33 - 17) + (33 - 22)} \times 10$$

$$\approx 20.909$$

Modal class = the class of 20-30

$$L_o = 20, f_o = 33, f_{-1} = 17, f_1 = 22,$$

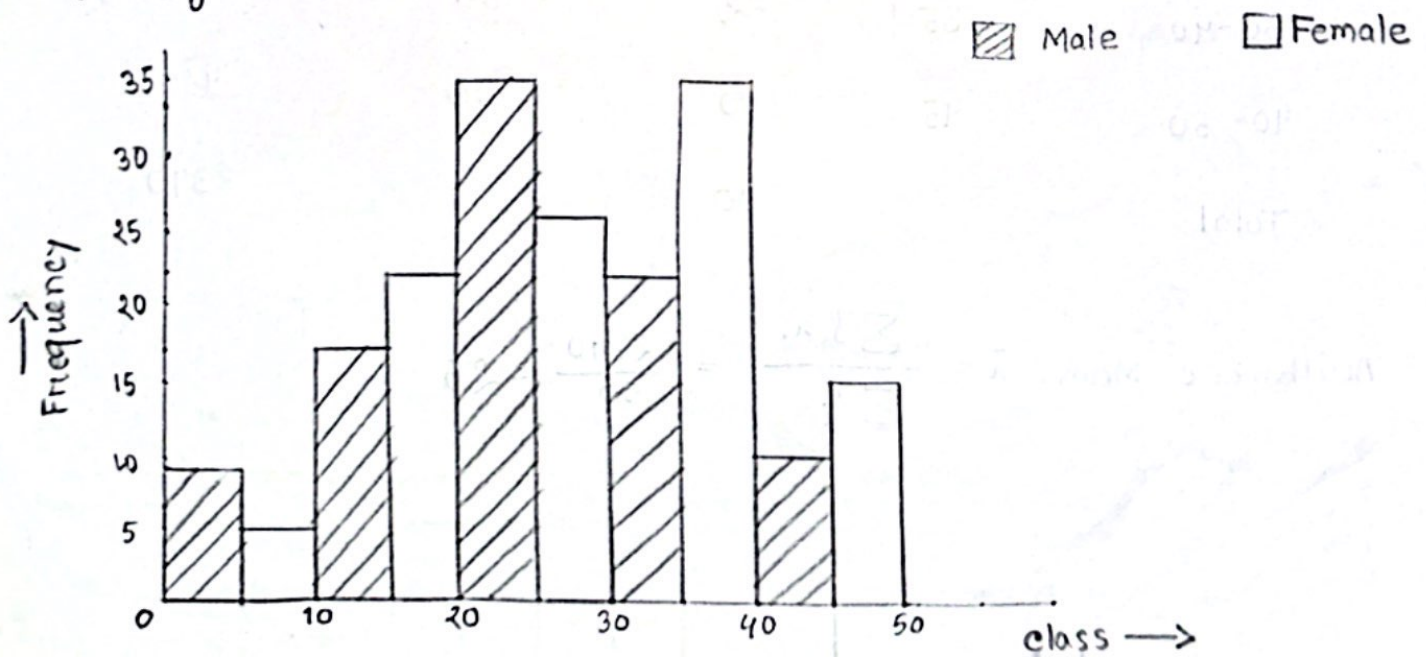
$$C = 10$$

(c) From "b" it can be seen that,

$$\text{Mean} > \text{Median} > \text{Mode}$$

So, the shape of the distribution of marks for male students is positively skewed.

(d) Histogram would be more appropriate:



Student id: 24101226, $x = 6$

Screen Brightness(%)	16	19	37	40	61	53	32	62	67	59
Battery Life (Hours)	20	21	16	18	16	9	19	16	8	6

(a) $E(y) = \beta_0 + \beta_1 x$

$\hat{y} = b_0 + b_1 x$

$\therefore \hat{y} = 24.712 - 0.22x$

$\therefore \text{Battery Life(hours)} = 24.712 - 0.22x$

$b_1 = \frac{\sum x_i y_i - n \bar{x} \bar{y}}{\sum x_i^2 - n \bar{x}^2}$

$b_1 = \frac{5974 - 10 \times 44.6 \times 14.9}{22954 - 10 \times 44.6^2}$

$= -0.22$

$\bar{x} = 44.6$

$\sum x_i^2 = 22954$

$n = 10$

$\bar{y} = 14.9$

$\sum x_i y_i = 5974$

$b_0 = \bar{y} - b_1 \bar{x}$

$= 14.9 - (-0.22) \times 44.6$

$= 24.712$

(b) For 27%,

$\text{Battery Life(hours)} = 24.712 - 0.22 \times 27$

$= 18.772 \text{ hours}$

(Answer)

(c) Residual, $e_i = y_i - \hat{y}_i$

(For 40%)

$= 18 - 15.912$

$= 2.088$

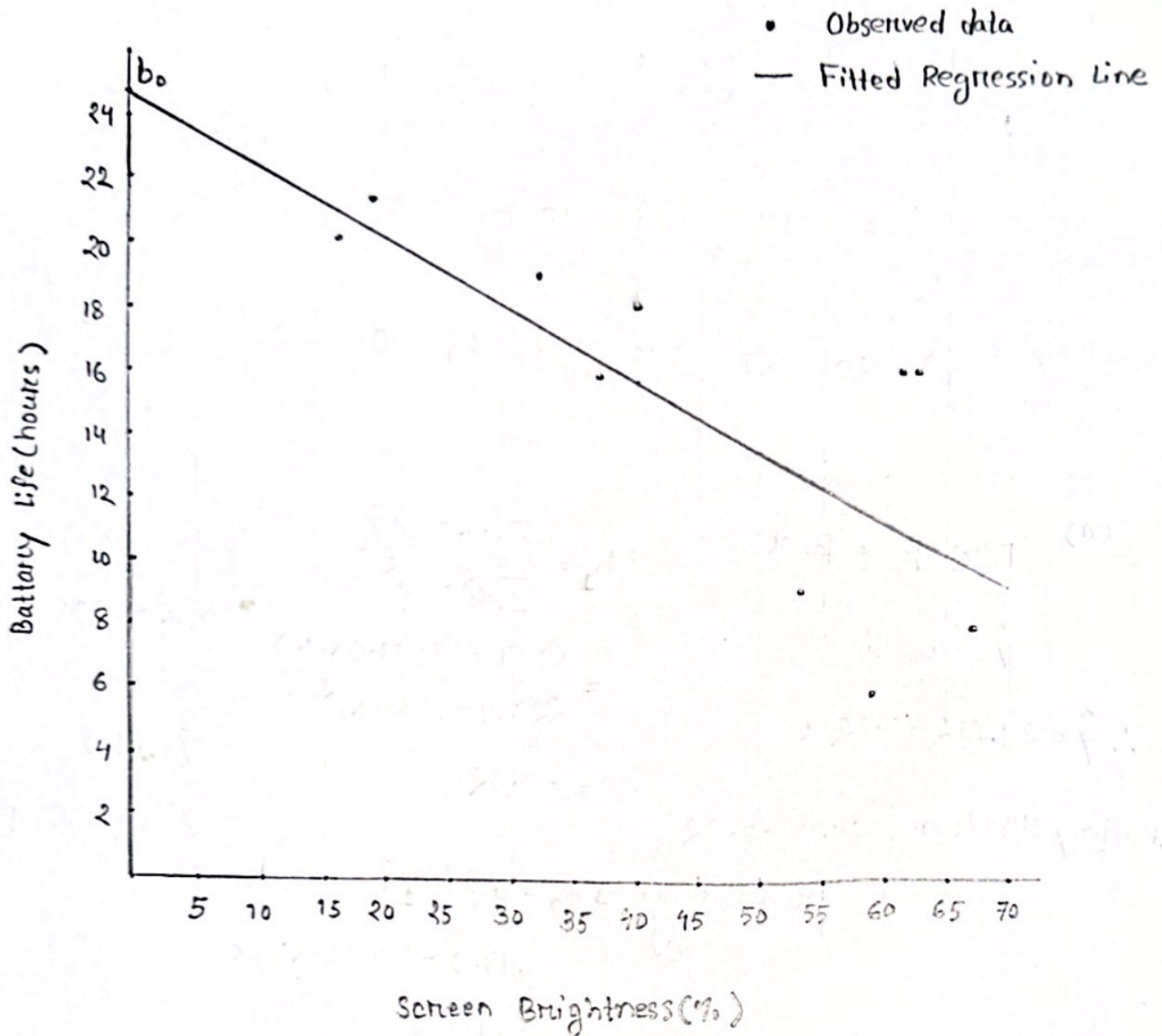
$y_i = 18$

$\hat{y}_i = 24.712 - 0.22 \times 40$

$[x=40]$

$= 15.912$

(d)



From the model we get that the slope of the regression line is $-2 - 0.22$.

We know,

$$r = \frac{\sum xy - n\bar{x}\bar{y}}{\sqrt{(\sum x^2 - n\bar{x}^2)(\sum y^2 - n\bar{y}^2)}}$$

$$= \frac{5975 - 10 \times 44.6 \times 14.9}{\sqrt{(22957 - 10 \times 44.6^2)(2475 - 10 \times 14.9^2)}}$$

$$= -0.7584$$

$$\sum xy = 5975$$

$$\bar{x} = 44.6$$

$$\bar{y} = 14.9$$

$$n = 10$$

$$\sum x^2 = 22954$$

$$\sum y^2 = 2475$$

$$\therefore r^2 = 0.575$$

As well as

Graphically and as well as mathematically it is shown in the answer script that Battery Life(^{hours}%) is related to Screen Brightness(%) moderately strong in negative direction.

(e) From "d", $r^2 = 0.575$

The value of r^2 indicates that the model explains 57.7% of the variation in battery life, which is moderate fit.

Answer to the Question No-3

Student Id : 24101226 , X = 2

Dhawan	25	51	40	77	52	36	92	83	40	37
Gill	39	59	86	27	122	42	104	36	88	34

(a) Top 10% means the score above 90%.

here, $n = 10$

$$\frac{i \times n}{100} = \frac{90 \times 10}{100} = 9, \text{ an integer}$$

Arranging Gill's score:

27, 34, 36, 39, 42, 59, 86, 88, 104, 122

$$\therefore 90^{\text{th}} \text{ Percentile, } P_{90} = \frac{1}{2} \left[\left(\frac{i \times n}{100} \right)^{\text{th}} \text{ value} + \left(\frac{i \times n}{100} + 1 \right)^{\text{th}} \text{ value} \right]$$

$$= \frac{1}{2} (9^{\text{th}} \text{ value} + 10^{\text{th}} \text{ value})$$

(a) Arranging Gill's score

27, 34, 36, 39, 42, 59, 86, 88, 104, 122

Top 10% means the score above 90%.

here 90th Percentile score is 104.

∴ The score above which to 10% score belong for Shubane

Shubman Gill is 104.

(b) After ~~arranging~~

~~Dhawan~~ For Dhawan,

$$\text{Average score} = \frac{25+51+40+77+52+36+92+83+40+37}{10}$$

$$= 53.3$$

$$\bar{x} = 53.3$$

$$\text{Standard deviation} = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2}$$

$$= \sqrt{\frac{(25-53.3)^2 + (51-53.3)^2 + (40-53.3)^2 + \dots}{10}}$$

$$= \frac{1}{10} \sqrt{(25-53.3)^2 + (51-53.3)^2 + (40-53.3)^2 + (77-53.3)^2 + (52-53.3)^2 + (36-53.3)^2 + (92-53.3)^2 + (83-53.3)^2 + (40-53.3)^2 + (37-53.3)^2}$$

$$= 22.7 \quad 21.6$$

For Gill,

$$\text{Average score} = \frac{27 + 34 + 36 + 39 + 42 + 59 + 86 + 88 + 104 + 122}{10}$$

$$= 63.7 = \bar{x}$$

Standard deviation

$$= \sqrt{\frac{(27 - \bar{x})^2 + (34 - \bar{x})^2 + (36 - \bar{x})^2 + (39 - \bar{x})^2 + (42 - \bar{x})^2 + (59 - \bar{x})^2 + (86 - \bar{x})^2 + (88 - \bar{x})^2 + (104 - \bar{x})^2 + (122 - \bar{x})^2}{10}}$$

$$= 31.95$$

~~As Dhawan has a low~~

~~Stu Gill~~ has a higher average score (63.7) compared to Dhawan (53.3), but Dhawan has a lower standard deviation (21.6) than Gill ~~(53.3)~~ (31.95), which means Dhawan is more consistent

$\therefore$ I would select Dhawan for his consistency.